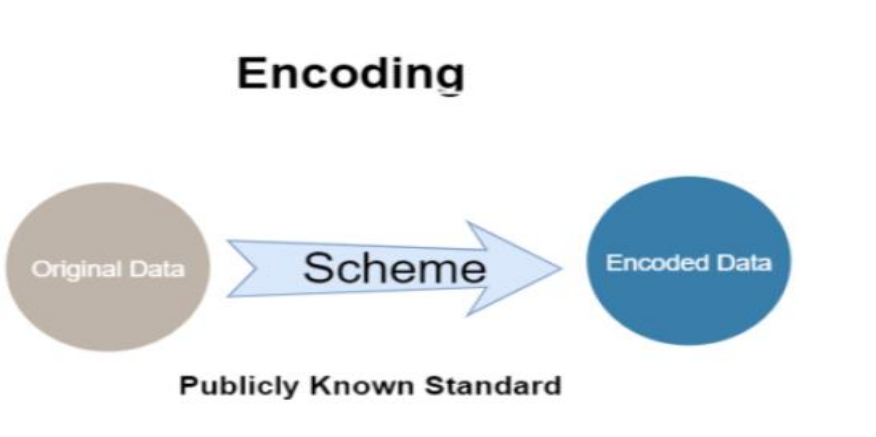
Encoding Schemes

(Module 1 .c)



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Objective

By the end of this class, students will be able to

- Define data encoding.
- Explain the purpose of encoding in web applications.
- Identify common encoding schemes.
- Differentiate encoding from encryption.
- Demonstrate basic encoding and decoding techniques.

Encoding Schemes in Web Security

Encoding is the process of converting data into a different format so that it can be safely transmitted, stored, or processed by web applications.

Examples

- URL Encoding
- Base64 Encoding
- HTML Encoding
- Unicode Encoding

ENCODING

Encoding converts information from one representation into another representation.

Example

Hello World

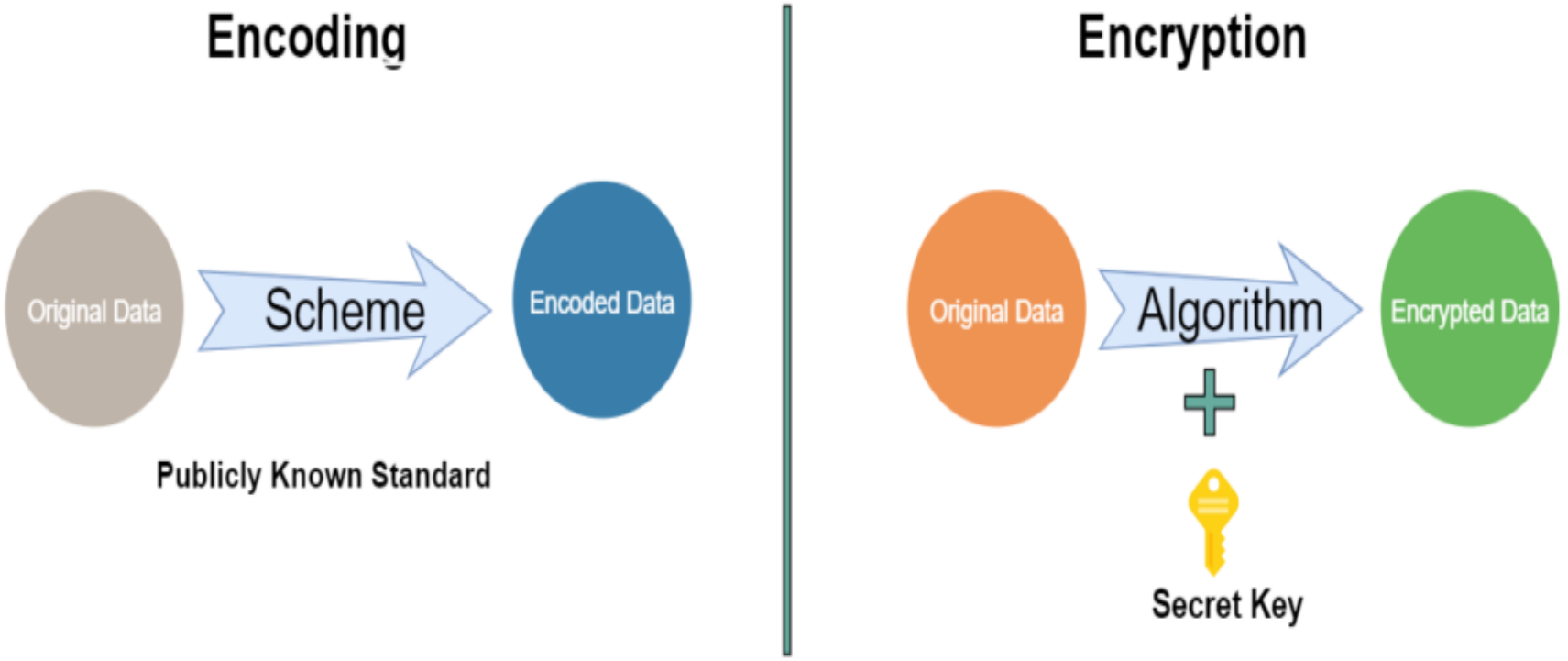
URL Encoded

Hello%20World

Important

Encoding does not provide confidentiality. It mainly ensures that data can be correctly handled or transmitted.

Encoding vs Encryption



Encoding ≠ Encryption

Encoding vs Encryption

Encoding	Encryption
Changes data format	Protects data confidentiality
Easily reversible	Requires a key for decryption
Used for compatibility and transmission	Used for security
Example: Base64	Example: AES

URL Encoding

URL encoding converts special characters into a format that can safely be transmitted in a URL.

Example

- **Hello World!**

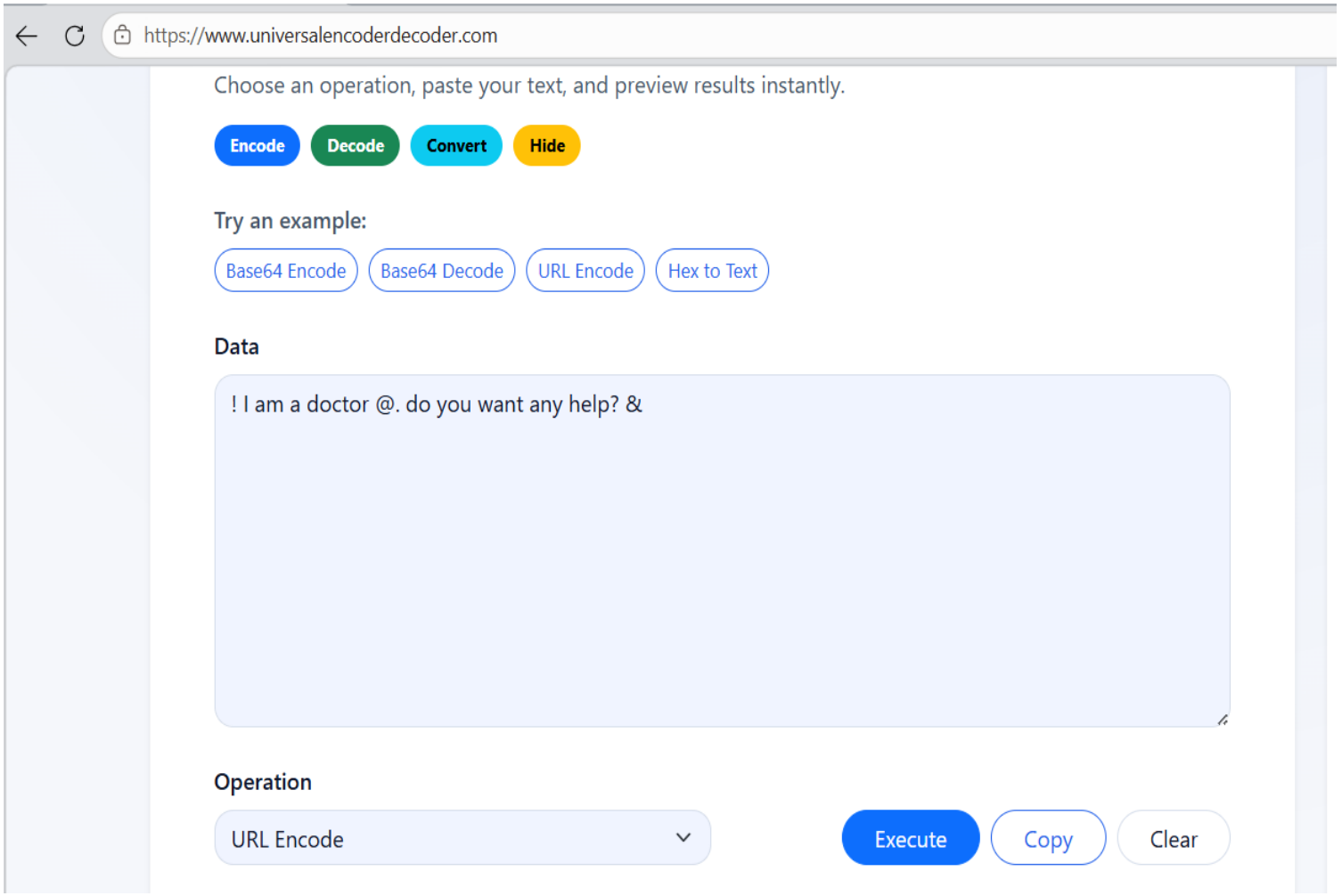
Becomes

- **Hello%20World%21**

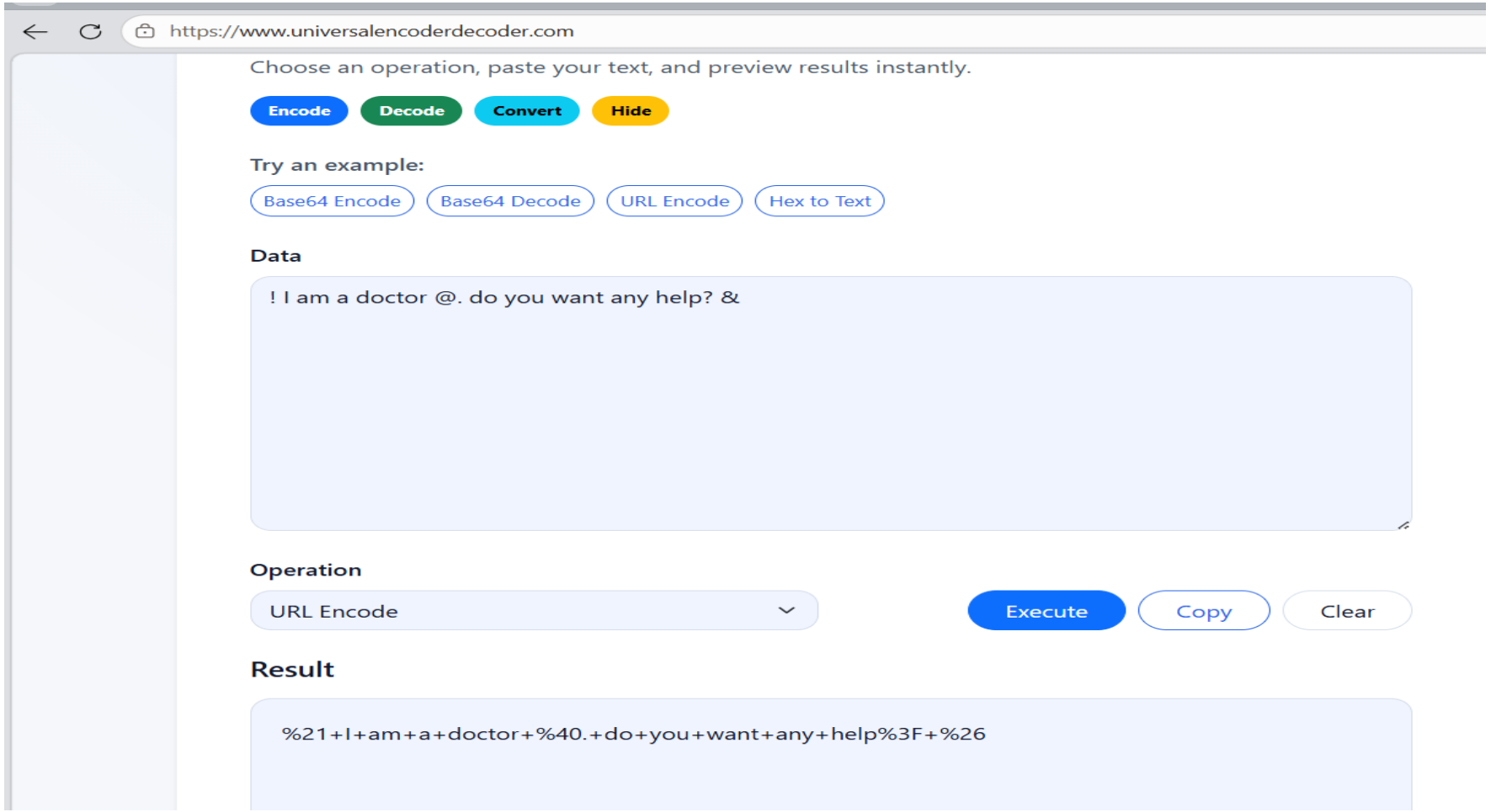
URL Encoding Example

Character	URL Encoding
Space	%20
!	%21
@	%40
&	%26

URL Encoding Demo



URL Encoding Demo



Base64 Encoding

Base64 converts binary data or text into ASCII characters.

Example

Text - Hello

Base64 -SGVsbG8=

Uses

- Email attachments
- Web APIs
- Data transmission
- Embedding images in web pages

Important

Base64 is not encryption and does not protect sensitive information.

Base64 Encoding Example Using ASCII

Step 1: Convert characters to ASCII

Character	ASCII (Decimal)	Binary (8-bit)
H	72	01001000
e	101	01100101
l	108	01101100
l	108	01101100
o	111	01101111

Base64 Encoding Example Using ASCII

Step 2: Combine the binary values

01001000 01100101 01101100 01101100 01101111

Step 3: Split into 6-bit groups

010010 000110 010101 101100 011011 000110 1111

Add padding zeros to complete the last group:

010010 000110 010101 101100 011011 000110 111100

Base64 Encoding Example Using ASCII

Step 4: Convert each 6-bit group to decimal

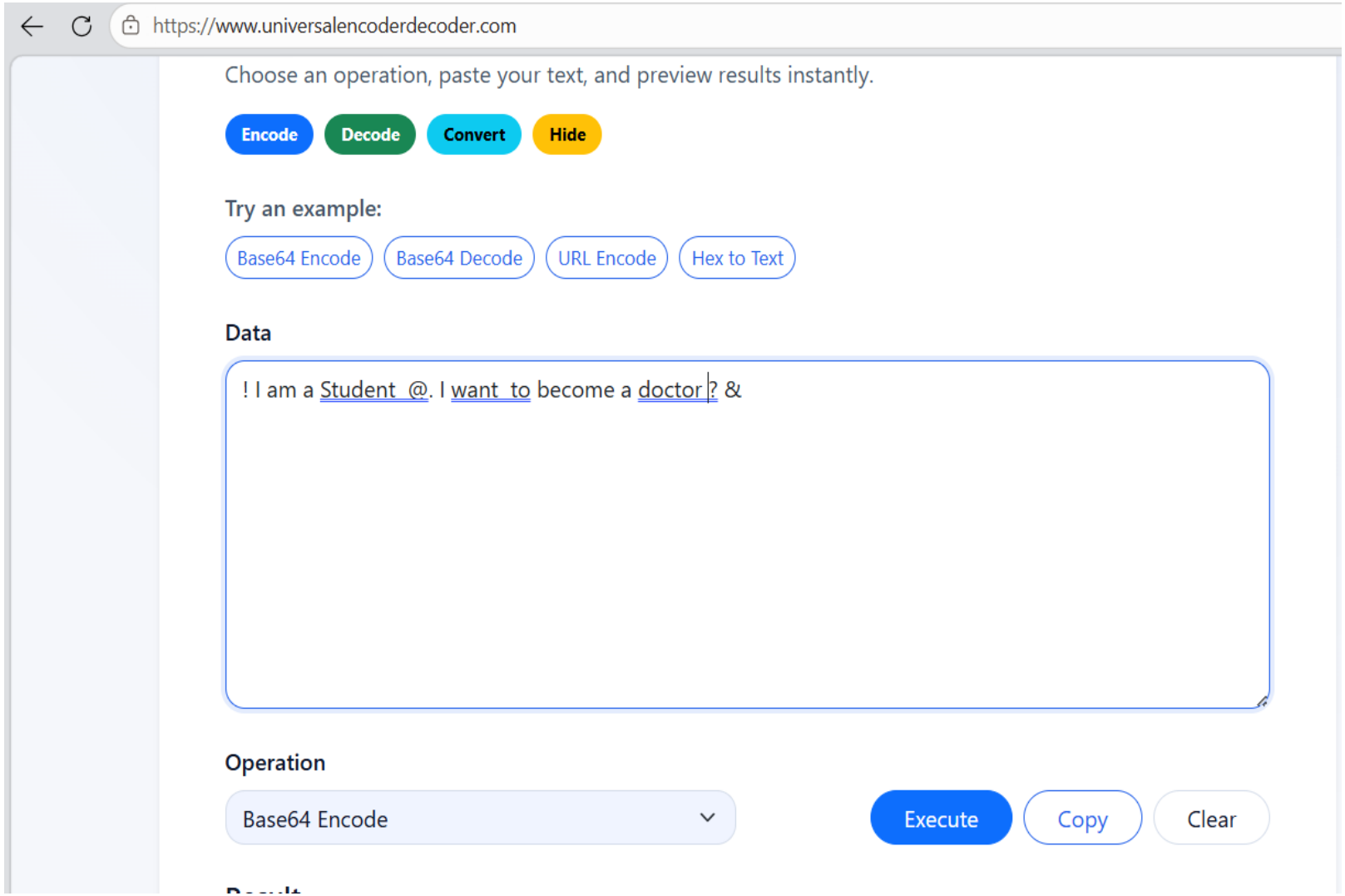
Binary	Decimal	Base64 Character
010010	18	S
000110	6	G
010101	21	V
101100	44	s
011011	27	b
000110	6	G
111100	60	8

Final Base64 Output

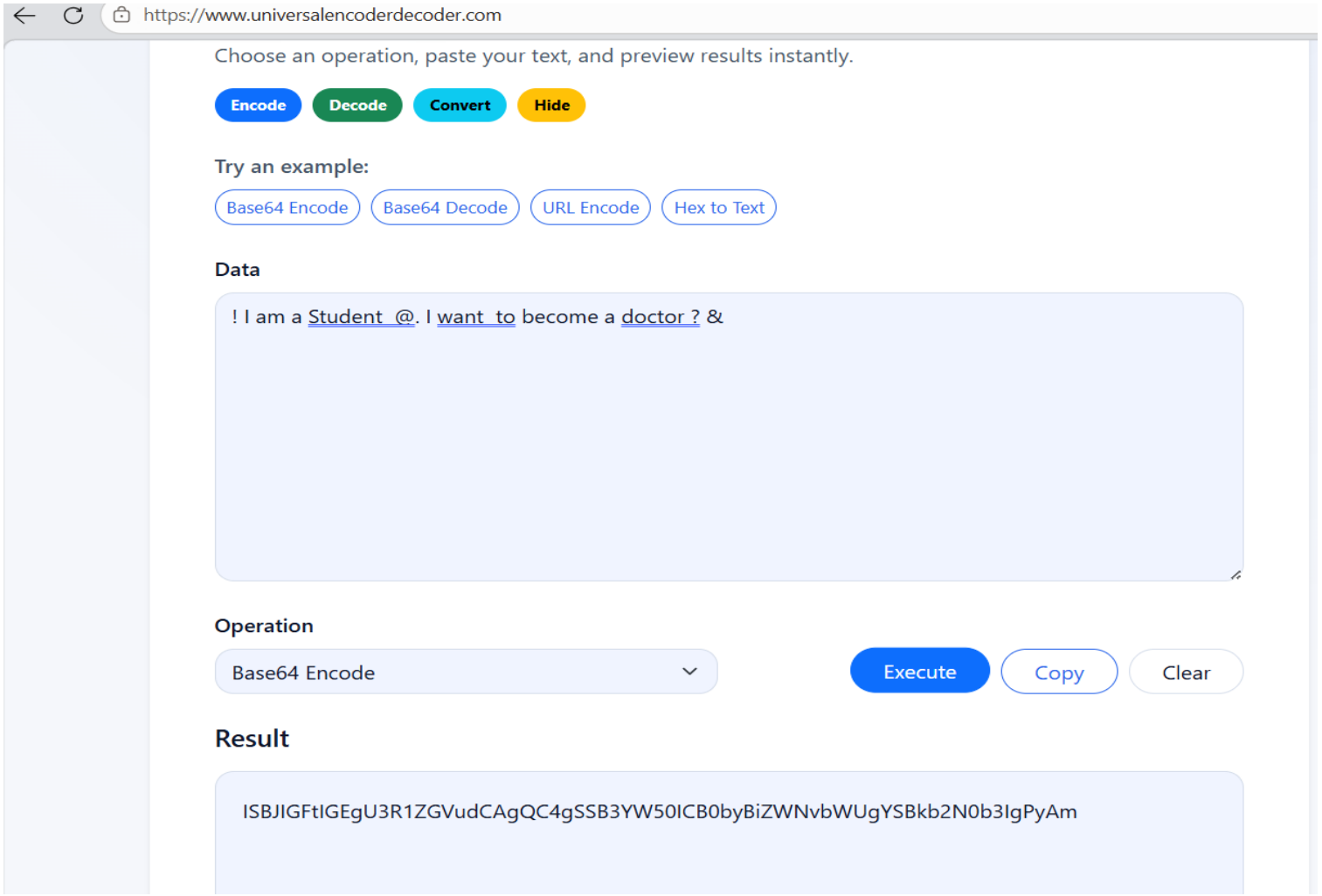
Hello → SGVsbG8=

Note: The = symbol is padding used to make the Base64 output length a multiple of 4.

Base64 Encoding Demo



Base64 Encoding Demo



HTML Encoding

HTML encoding converts special HTML characters into safe representations.

Example

Character	HTML Encoding
<	<
>	>
&	&
"	"

HTML Encoding

Example

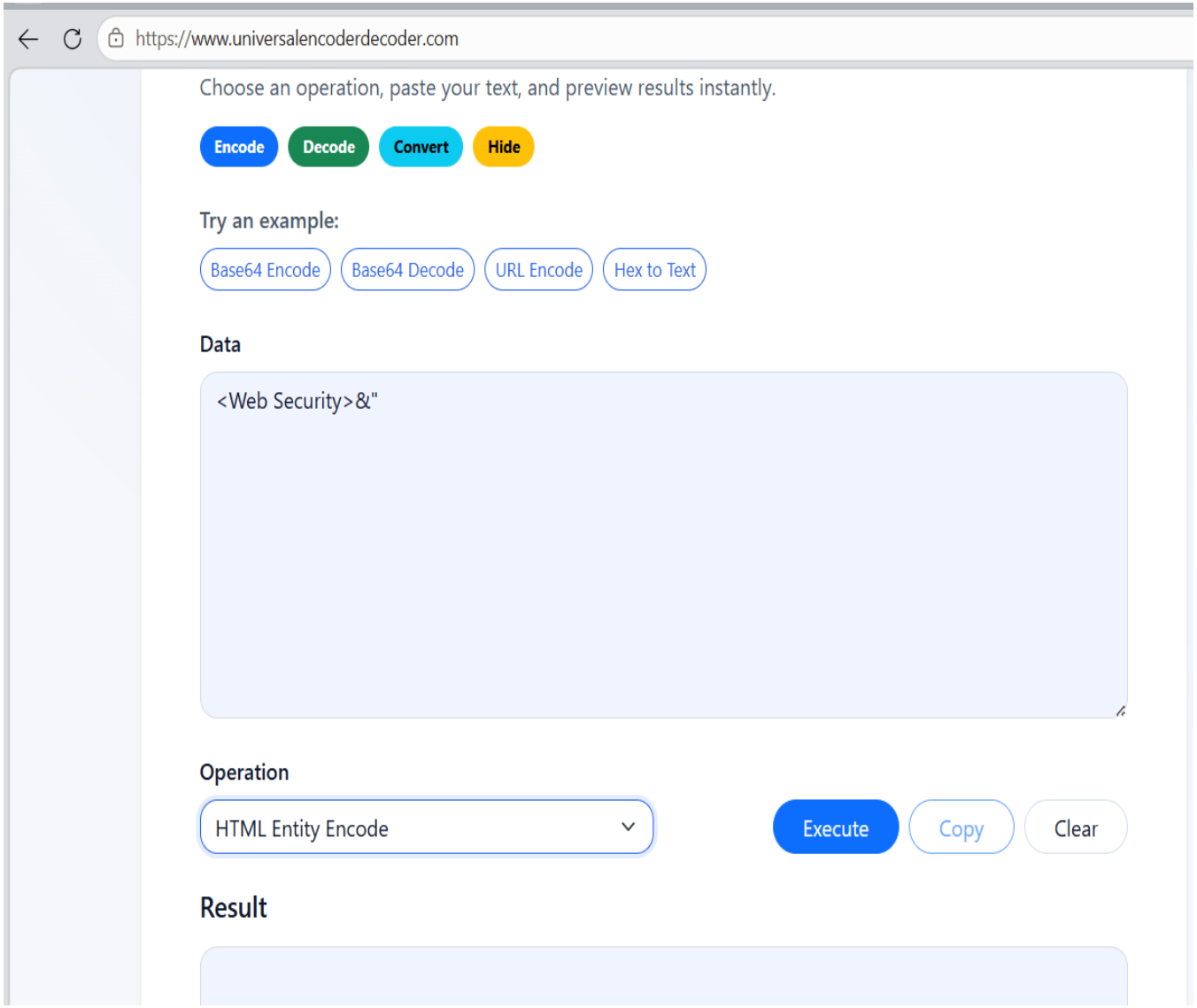
<script>

becomes

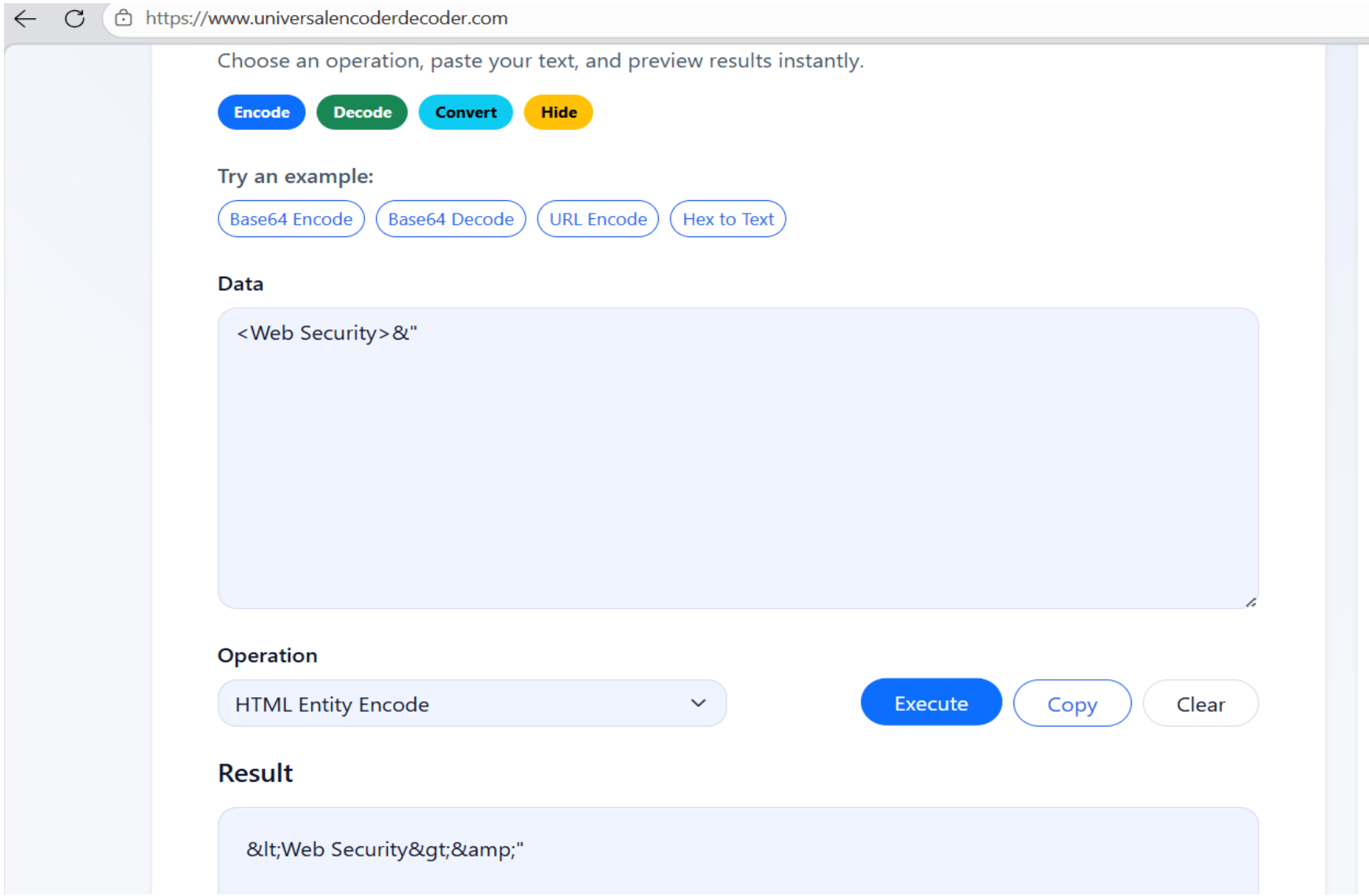
<script>

This helps browsers display special characters safely as text.

HTML Encoding Demo



HTML Encoding Demo



Unicode Encoding

Unicode encoding represents each character using a unique code point, written in the format U+XXXX. This allows computers to represent and process characters from different languages, symbols, and emojis.

- **UTF-8** is one of the most commonly used character encoding formats on the web.

Example:

café ⟶ U+0063 U+0061 U+0066 U+00E9

Unicode Encoding

For example

- c has decimal value **99**
- Decimal **99** = Hexadecimal **63**
- Therefore, Unicode represents it as **U+0063**

Similarly

- a → Decimal **97** → Hexadecimal **61** → **U+0061**
- f → Decimal **102** → Hexadecimal **66** → **U+0066**

Why are zeros added?

- Unicode code points are commonly displayed with leading zeros for a consistent format:
- **63 → 0063**

Unicode Encoding

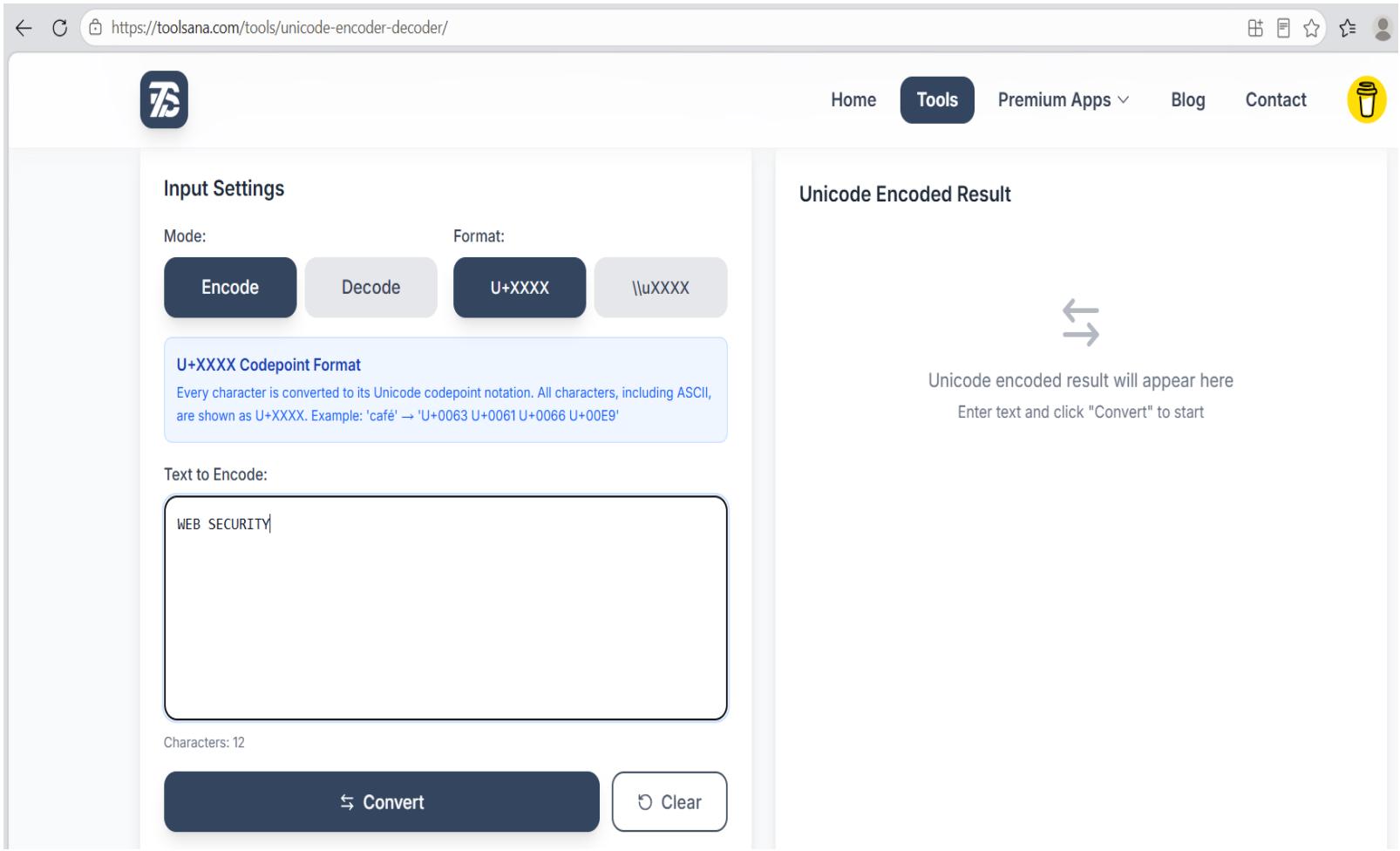
Example: café

Character	Unicode Code Point	Hexadecimal Value
c	U+0063	0063
a	U+0061	0061
f	U+0066	0066
é	U+00E9	00E9

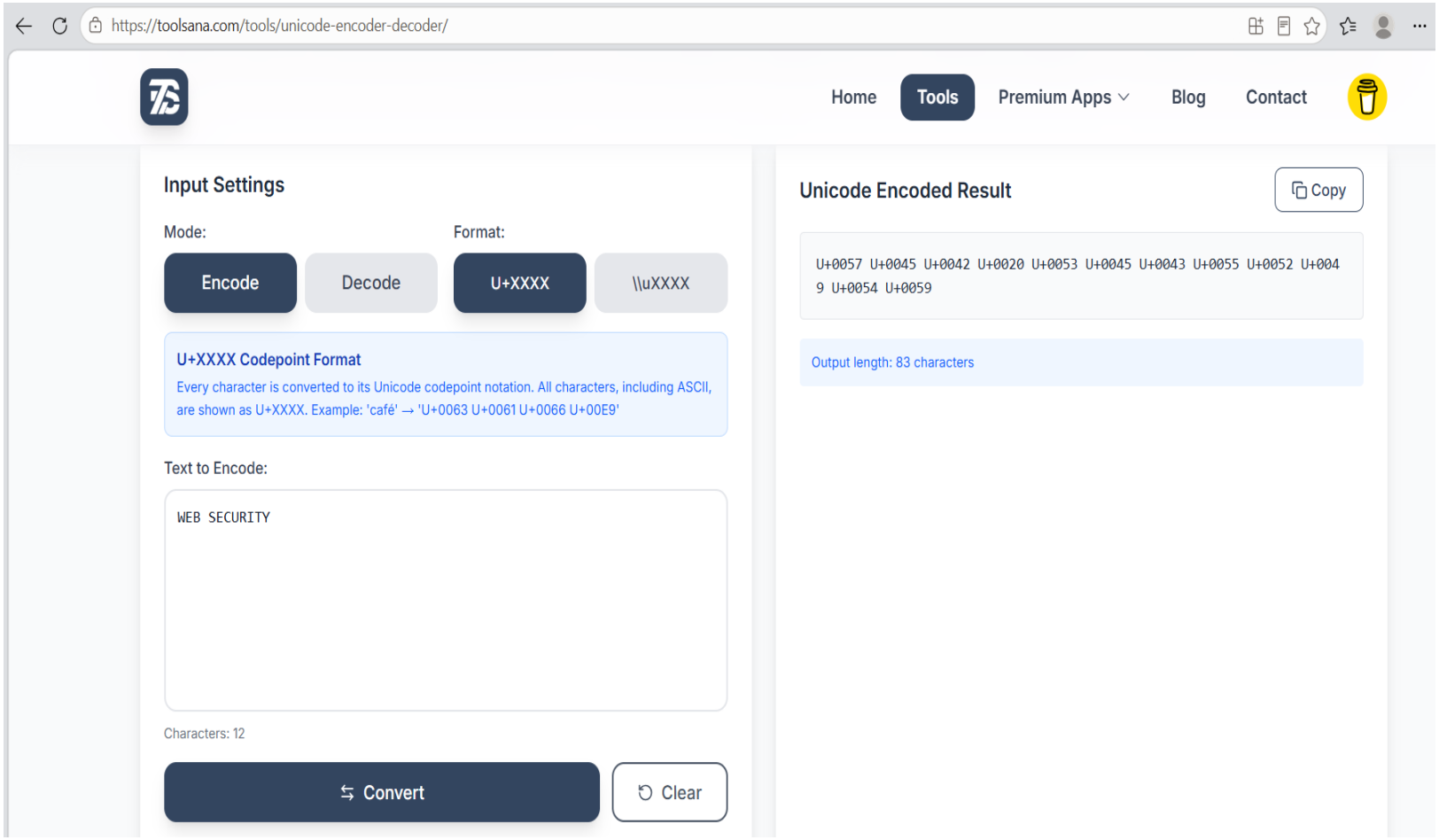
Unicode gives every character a unique number. This number is usually written in hexadecimal format and displayed as U+XXXX

.

Unicode Encoding Demo



Unicode Encoding Demo



Security Importance of Proper Encoding

- Proper encoding helps web applications
- Handle special characters correctly.
- Safely transmit data through URLs.
- Display text correctly in browsers.
- Support multiple languages.
- Reduce some input-handling security risks when appropriate output encoding is used.

Note: Encoding alone cannot secure passwords, confidential information, or sensitive data.

Classroom Activity 1 : MCQ

1. What is the main purpose of encoding?

A. To permanently delete data
B. To change data into another format
C. To encrypt confidential information
D. To block network traffic

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Classroom Activity 1 : MCQ

2. Which encoding scheme is commonly used for special characters in URLs?

A. Base64
B. AES
C. URL Encoding
D. SHA-256

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B. AES
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Classroom Activity 1 : MCQ

3. Which HTML encoding represents the < character?

A. %20

B. <

C. SGVsbG8=

D. #40

Classroom Activity 1 : MCQ

3. Which HTML encoding represents the < character?

A. %20

B. <

C. SGVsbG8=

D. #40

Summary

- Encoding converts data into another format.
- Common web encoding schemes include **URL Encoding, Base64, HTML Encoding, and Unicode**.
- Encoding and encryption are different.
- Encoding does not provide confidentiality.
- Proper output encoding is important for secure web application development.

Thank you